

Do Emotions in Prompts Matter? Effects of Emotional Framing on Large Language Models

Minda Zhao^{1†}, Yutong Yang^{1†}, Chufei Peng^{1†},
Rachel Gonsalves^{1†}, Weiyue Li^{1†}, Ruyi Yang¹, Zhixi Liu²,
Mengyu Wang^{1*}

¹Harvard University.

²Bryn Mawr College.

*Corresponding author(s). E-mail(s): mengyu-wang@meei.harvard.edu;

Contributing authors: mindazhao@hsph.harvard.edu;
yutongyang@hsph.harvard.edu; chufeipeng@g.harvard.edu;
rgonsalves@mba2027.hbs.edu; weiyueli@fas.harvard.edu;
ruyi-yang@gsd.harvard.edu; liu2@brynmawr.edu;

[†]These authors contributed equally to this work.

Abstract

Emotional tone is pervasive in human communication, yet its influence on large language model (LLM) behaviour remains unclear. Here, we examine how first-person emotional framing in user-side queries affect LLM performance across six benchmark domains, including mathematical reasoning, medical question answering, reading comprehension, commonsense reasoning and social inference. Across models and tasks, static emotional prefixes usually produce only small changes in accuracy, suggesting that affective phrasing is typically a mild perturbation rather than a reliable general-purpose intervention. This stability is not uniform: effects are more variable in socially grounded tasks, where emotional context more plausibly interacts with interpersonal reasoning. Additional analyses show that stronger emotional wording induces only modest extra change, and that human-written prefixes reproduce the same qualitative pattern as LLM-generated ones. We then introduce EmotionRL, an adaptive emotional prompting framework that selects emotional framing adaptively for each query. Although no single emotion is consistently beneficial, adaptive selection yields more reliable gains than fixed emotional prompting. Together, these findings show that emotional tone is neither a dominant driver of LLM performance nor irrelevant noise, but a weak and input-dependent signal that can be exploited through adaptive control.

1 Introduction

Large language models (LLMs) have become ubiquitous across domains including education, coding assistance, creative writing, and personal productivity [1–3]. As LLM-generated text increasingly appears in consumer communications, corporate messaging, and international correspondence [1], understanding how these systems process and respond to human communication has become critical. LLMs are trained on large-scale human language corpora and learn statistical patterns of human communication, yet they do not operate as neutral systems [4]. This recognition has driven extensive research into LLM behaviors, including hallucination, fairness, toxicity, sycophancy, and consistency [5].

Human communication is inherently emotional. Individuals convey affect through tone, framing, and lexical choice when expressing needs, negotiating disagreements, or seeking advice [6, 7]. Consequently, real-world human–LLM interactions often carry an emotional tone that, even when unstated, shapes model outputs and influences how users interpret responses [8]. Emotion is therefore central rather than peripheral to human–LLM interaction.

Emotion plays a causal role in human decision-making. Emotional states guide attentional focus, bias interpretation under ambiguity, and influence reasoning in uncertain environments [9, 10]. These effects can alter behavior even when material incentives remain unchanged. Empirical work demonstrates that emotion functions as a causal mechanism linking context to behavior. For example, Heffner et al. [11] studied emotion prediction errors in social decision-making tasks involving punishment and forgiveness. They showed that discrepancies between expected and experienced emotion predicted subsequent decisions beyond reward-based prediction errors, isolating a pathway through which emotion updates future behavior and supports adaptive social interaction. Related work has shown that emotion-guided attention impacts perceptual decision-making [12], emotional expressions influence visual search efficiency [13], and affective information integrates into experience, judgment, and decision-making [14]. Because LLMs are trained on emotionally mediated human text, affective tone in prompts may similarly modulate model behavior. However, the causal mechanisms through which emotional cues influence LLM performance remain underexplored.

Prior work has primarily studied emotion in LLMs through explicit prompt manipulations. In strategic settings, assigning emotional roles or varying task framing has been shown to shift model behavior in cooperation, bargaining, and related game scenarios [15, 16]. Related work has further shown that emotionally charged descriptions can alter norm-enforcement judgments, leading models to impose stronger punishment in altruistic punishment settings [17]. Beyond strategic decision-making, several studies have reported that affective prompts can influence downstream task performance, including EmotionPrompt-style approaches designed to improve results across benchmarks [18–20]. Other lines of work have examined emotional expression in LLMs, responses to conflicting emotional cues, controlled affective text generation, and fine-grained emotion resources [21–24]. Taken together, these studies show that LLMs are responsive to explicitly inserted emotional content. However, three important questions remain unresolved. First, it is still unclear whether emotionally framed prompts expressed in a first-person, user-voiced form produce systematic and reliable changes

in model behavior across standard tasks. Second, prior work has not directly tested whether these effects depend on prompt authorship, particularly whether they hold for human-written emotional prefixes as well as LLM-generated ones. Third, existing studies have largely treated emotion as a fixed prompting condition rather than learning how emotional framing should be selected for a given input. These questions are particularly important given documented biases in LLMs [25–27], including cognitive biases in clinical settings [28, 29], stigmatizing language in medical records [30, 31], and fairness concerns [32].

In this work, we study how first-person emotional framing in user prompts affects LLM behavior. Rather than assigning emotions to the model or relying on coarse positive-negative distinctions, we prepend short first-person emotional prefixes that express the user’s affective state and examine six emotion categories: happiness, surprise, fear, sadness, disgust, and anger, guided by Plutchik’s basic emotions framework [33] and Russell’s circumplex model of affect [34]. We study these effects across reasoning and knowledge-based tasks, focusing on how task structure and complexity shape emotional modulation. Our contributions are threefold. First, we show that static emotional prompting is not a reliable general-purpose intervention: across benchmarks, fixed emotional prefixes typically induce only small average accuracy shifts, with effects that vary substantially across tasks, backbone models, and emotion categories rather than following a uniform pattern. Second, we demonstrate that this overall pattern is robust to targeted validation studies: increasing emotional intensity produces only modest additional change without creating a qualitatively different regime, and human-written prefixes reproduce the same qualitative conclusions as LLM-generated ones. Third, we introduce EmotionRL, an adaptive emotional prompting framework that formulates emotional prompting as an input-conditioned decision problem, and show that different questions benefit from different emotional framings. By selecting a more suitable emotion for each input, EmotionRL yields more reliable gains than either the no-emotion baseline or the average of fixed emotional prompts.